VinayMattapalli / **demand-forecasting-inventory-system**

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README

Vinay Mattapalli

Senior AI/ML Engineer with 8+ years of experience designing and deploying production-grade machine learning and Generative AI systems across fintech and retail domains.

Expertise in real-time ML pipelines, fraud detection, and scalable microservices using Python, FastAPI, and AWS. Strong experience in RAG systems, multi-agent architectures, and MLOps with Docker, Kubernetes, and CI/CD pipelines.

Technical Skills

Programming: Python, SQL Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM Generative AI: RAG, LangChain, LangGraph, Llama, Transformers, Prompt Engineering Data Engineering: Apache Spark, Kafka, Airflow MLOps: MLflow, CI/CD, Docker, Kubernetes, Terraform Cloud: AWS (EKS, Lambda, S3, API Gateway, RDS) Vector Databases: FAISS, Pinecone, pgvector Monitoring: Prometheus, Grafana

Professional Experience

Senior AI/ML Engineer – Stripe (2024 – Present)

- Built real-time fraud detection and risk scoring systems using XGBoost and PyTorch
- Designed RAG-based AI systems for fraud investigation using LangChain and LangGraph
- Developed FastAPI microservices with sub-150ms latency
- Built Kafka + Spark pipelines for real-time feature processing
- Implemented vector search using FAISS and Pinecone
- Deployed scalable systems using Docker, Kubernetes, and AWS

ML Engineer – Dollar General (2022 – 2024)

- Built demand forecasting models using XGBoost and time-series techniques
- Improved forecast accuracy and reduced stockouts across retail systems
- Developed ETL pipelines using PySpark and SQL
- Deployed ML models using AWS services

Data Engineer – Aisera (2020 – 2022)

- Built data pipelines for NLP and chatbot systems
- Developed ETL workflows and integrated enterprise APIs
- Supported AI model training and deployment

Featured Projects

Mine or Rock Detection

- Logistic Regression model for sonar signal classification
- Achieved 88% accuracy with feature engineering and tuning

Diabetes Prediction System

- Built ML models using SVM and Random Forest
- Improved accuracy through feature scaling and tuning

RAG Chatbot System

- Built Retrieval-Augmented Generation system using FAISS and LangChain
- Enabled intelligent document-based querying

Contact

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Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

VinayMattapalli Vinay Mattapalli

Languages

- **Python** 99.0%
- **Dockerfile** 1.0%

Suggested workflows

Based on your tech stack

SLSA Generic generator

Generate SLSA3 provenance for your existing release workflows

Configure

Django

Build and Test a Django Project

Configure

Python application

Create and test a Python application.

Configure

More workflows

Dismiss suggestions